

Machine Learning Application to Assess Occurrence and Saturations of Methane Hydrate in Marine Deposits Offshore India

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Abstract

Artificial Neural Networks (ANN) were used to assess methane hydrate occurrence and saturation in marine sediments offshore India. The ANN analysis classifies the gas hydrate occurrence into three types: methane hydrate in pore space, methane hydrate in fractures, or no methane hydrate. Further, predicted saturation characterizes the volume of gas hydrate with respect to the available void volume. Log data collected at six wells, which were drilled during the India National Gas Hydrate Program Expedition 02 (NGHP-02), provided a combination of well log measurements that were used as input for machine learning (ML) models. Well log measurements included density, porosity, electrical resistivity, natural gamma radiation, and acoustic wave velocity. Combinations of well logs used in the ML models provide good overall balanced accuracy (0.79 to 0.86) for the prediction of the gas hydrate occurrence and good accuracy (0.68 to 0.92) for methane hydrate saturation prediction in the marine accumulations against reference data. The accuracy scores indicate that the ML models can successfully predict reservoir characteristics for marine methane hydrate deposits. The results indicate that the ML models can either augment physics-driven methods for assessing the occurrence and saturation of methane hydrate deposits or serve as an independent predictive tool for those characteristics.

Keywords: artificial neural network; methane hydrate; well log data

Introduction

Gas hydrates are solids containing water and gas, where included gas is typically methane (throughout the paper “gas hydrate” and “methane hydrate” are used interchangeably). Methane hydrate is poised as a promising substance for future energy demands (Boswell et al., 2014; Ruppel and Kessler, 2017). Gas hydrate naturally occurs in areas within and beneath permafrost (Boswell et al., 2014; Collett et al., 2011) and marine settings (Collett et al., 2019; Konno et al., 2017). The total resource potential of marine gas hydrate accumulations exceeds that of permafrost-associated deposits (Boswell et al., 2014; Sahu et al., 2020). A number of techniques can be employed to detect gas hydrates including seismic analyses (Boswell et al., 2016; Plaza-Faverola et al., 2017), coring (Boswell et al., 2019; Dai and Santamarina, 2017; Merey, 2016; Riedel et al., 2010; Yoneda et al., 2021), and well logging (Riedel et al., 2010). At a methane hydrate research drill site, field testing logging-while-drilling (LWD) data are often acquired to improve the quality of the log data relative to more traditional wireline conveyed downhole logging tools (Merey, 2016). During LWD acquisition, numerous well log measurements are made including electrical resistivity, compressional and shear acoustic velocities, gamma-ray, and nuclear magnetic resonance (NMR). Due to the unconsolidated nature of marine sediments, borehole washout often represents a problem relative to the quality of the acquired downhole log data (Merey, 2019). To improve the quality of log reading a method of compensating for the effect of washouts can be used that implies modeling the washed-out portion of the borehole as a vertical layer filled with seawater (drilling fluid) (Lee, M. W., 2012).

The estimation of methane hydrate saturation (S_{mh}) delivers a quantitative assessment of methane hydrate reservoir properties and a location of prospective reservoir sections that have

been targeted for wellbore completion and induced depressurization testing (Xia et al., 2020). S_{mh} can be estimated from downhole log data using three methods based on direct measurements and empirical derivations: (1) electrical resistivity logs using the Archie relationship (Archie, 1942), (2) analysis of compressional- and shear-wave acoustic velocities (Lu and McMechan, 2002), and (3) a combination NMR and density well log analyses that consider NMR and density derived porosities and pore-fluid properties (Collett and Lee, 2012). Machine learning (ML) was recently used to derive petrophysical reservoir properties using log data (Ahmadi and Chen, 2019; Mahmoud et al., 2017; Moussa et al., 2018). In particular, ML was also engaged to characterize methane hydrate deposit properties (Ahmadi et al., 2020). In previous studies, various ML algorithms have been tested for lithological classification and S_{mh} estimation in methane hydrate accumulations using various well log combinations (WLC) (Chong et al., 2022; Singh et al., 2021, 2020).

Gas hydrates in marine settings often reside in sediment pores or fractures that can be further characterized with well log readings. Particularly, electrical resistivity of pore-filling hydrates demonstrates isotropic properties, while in the case of fracture-filling hydrates, resistivity log measurements are sensitive to fracture orientations relative to a downhole logging tool (Collett and Lee, 2012). Gas hydrates are characterized with high electrical resistivity. Exploiting this, resistivity-at-bit (RAB) electrical images are utilized to detect pore-filling and fracture-filling gas hydrates in sediments (Collett et al., 2019; Boswell et al., 2019). The various LWD tools used to survey gas hydrate deposits make measurements at different vertical resolutions and depths of investigation (see, for example, Table 2 in Collett et al., 2019 and

Boswell et al., 2017). Consequently, the quality of the well-log derived methane hydrate assessment depends on the quality of certain well log measurements, including electrical resistivity (Archie, 1942), NMR and density-derived porosities (Jain et al., 2019; Collett and Lee, 2012), and compressional and shear velocities (Lu and McMechan, 2002)). The specifics of the physical nature of the gas hydrate occurrence (i.e., interbedded sequences or more massive occurrence, and fracture- or pore-filling) also affect the well log-derived measurements (Collett et al., 2019; Boswell et al., 2014). Combinations of various well logs utilized in the ML approach allows mitigation of possible shortcomings for certain specific well logs (for example, in terms of the logging tool vertical resolution being comparable with thickness of gas hydrate-bearing and associated mud/shale subunits) and provide reliable estimates of the properties of interest. Gas hydrate occurrence and saturation (i.e., percent of sediment void space occupied by gas hydrate) interpreted using the physics-driven methods can serve as a ground truth, in other words, a reference data that allows the use of a supervised ML approach in the form of classification and regression. Classification refers to the predictive modeling of discrete labels, whereas regression is the continuous value prediction of variables. In this study, supervised ML serves as the main technique because the interpretation of the nature of the gas hydrate occurrence as derived from wellbore resistivity image analysis becomes the ground truth for classification, while the gas hydrate saturations derived wellbore resistivity logging becomes the ground truth for the regression derived values. In this work we apply the ML approach to well logs collected in offshore India to predict gas hydrate occurrence and saturation.

India's National Gas Hydrate Program Expedition 02 (NGHP-02) acquired a large amount of geophysical data including LWD, coring, and wireline logging data collected offshore India (Collett et al., 2019; Holland et al., 2019; Yadav et al., 2019; Hsiung et al., 2019; Nanda et al., 2019;). This study focused on downhole logging data acquired the following six holes: NGHP-02-05A, -02-06A, -02-07A, -02-08A, -02-09A, and -02-10A; that were established at the following six sites: Sites NGHP-02-05, -06, 07, -08, -09, and -10 in Area C as defined during the NGHP-02 expedition in the Krishna-Godavari Basin (Figure 1). Sites NGHP-02-07 and -10 along with Sites NGHP-02-08 and -09 are clustered together near the same location and geologic setting, while Sites NGHP-02-05 and NGHP-02-06 are located farther to the east at deeper water depths and near the base of the continental slope.

Table 1 presents information derived from downhole well log analysis (Collett et al., 2019) of methane hydrate deposits. These analyses confirmed the methane hydrate occurrence across the eastern edge of the continental crust of India (Yadav et al., 2019; Boswell et al., 2019; Collett et al., 2019). Pore-filling methane hydrate is found predominantly in coarse-grained, sandy-rich sedimentary sections, while fine-grained, clay-rich and mud-rich intervals more often host fracture-filling methane hydrate (Boswell et al., 2014). Typically, coarse-grained sandy-rich sediment provides a high-quality reservoir which results in relatively high gas hydrate saturation (70-90%) (Boswell et al., 2019; Collett et al., 2019). In Area C, the absence of adequate methane sources to fill the reservoirs limits the methane hydrate saturations at Sites NGHP-02-05 and -06 (Waite et al., 2019). The analysis of downhole log and core sample data also concluded that

limited gas hydrate occurrence in the high porosity reservoir section supported insufficient methane availability (Waite et al., 2019).

Table 1

NGHP-02 Site description (Area C) and comments on methane hydrate saturations each site*

Site	Gas hydrate, mbsf	Dominant lithology	Gas hydrate occurrence	Comments
NGHP-02-05	3326-3333 3341-3349	sand-rich sand-rich	pore-fill pore-fill	S_{mh} 20-70%; combination of a fully and partially gas charged system; insufficient gas to fully charge all available reservoirs (Collett et al., 2019)
NGHP-02-06	3093-3100 3128-3131	sand-rich sand-rich	pore-fill pore-fill	S_{mh} 40–90%
NGHP-02-07	2008-2042 2051-2054 2126-2147	mud-rich sand-rich mix	fracture-fill pore-fill fracture-fill	about 30% “underfilled” reservoir (Waite et al., 2019)
NGHP-02-08	2274-2297 2377-2380 2442-2458 2463-2468	mud-rich mud-rich sand-rich sand-rich	fracture-fill fracture-fill pore-fill pore-fill	$S_{mh} \sim 35\%$, $S_{mh} < 15\%$ 25 m with $> S_{gh}$ 80%
NGHP-02-09	2462-2490 2495-2512	sand-rich sand-rich	pore-fill pore-fill	48 m thick continuous S_{mh} nearly 80%
NGHP-02-10	1930-1971 1996-1999 2027-2033	mud-rich sand-rich mud-rich	fracture-fill pore-fill fracture-fill	S_{mh} about 20% S_{mh} about 50%

*The table assembled using information in (Collett et al., 2019).

Figure 2 presents the electrical resistivity well logs utilized to deduce methane hydrate saturation values (S_{mh}) by means of the Archie’s relationship (Archie, 1942; Collett and Lee, 2012) and the resistivity borehole image logs (RAB images acquired in Hole NGHP-02-08A as an example) used to determine the nature of methane hydrate occurrence in the corresponding reservoir sections (Collett et al., 2019). The higher resistivity log values (white color on the images) are interpreted as methane hydrate-bearing occurrence types. The initial occurrence assignment was reported in (Collett et al., 2019) using the structural analysis from RAB images which provided designation of reservoir intervals to certain occurrence types. The high-resolution electrical resistivity logs and the RAB images were further used to deduce S_{mh} and make occurrence assignments on a

point-to-point depth basis, respectively, that, in turn, become the ground truth in supervised ML training in this study.

The goal of this work is to employ ANN to differentiate methane hydrate-bearing and none methane hydrate-bearing sediments, classify gas hydrate occurrence and predict S_{mh} for marine gas hydrate deposits using downhole well log trained datasets for six wells drilled in offshore India. For that purpose of classifying the types of methane hydrate occurrence and assigning methane hydrate saturation values, classification and regression ML analyses were conducted.

Data Preprocessing and Machine Learning Method

The approach used in this study to predict S_{mh} was similar to previous work where ANN models were applied to characterize S_{mh} in permafrost methane hydrate deposits on the Alaska North Slope (USA) and in Northwest Territories (Canada) (Chong et al., 2022). In this previous study, ANN models were utilized to predict methane hydrate saturations in deposits. The approach for preparations of ML models as used in our study is given in Supplementary Materials associated with this paper (Figure S1). This approach was executed to build the ML models for methane hydrate saturation estimations and predict the nature of the methane hydrate occurrence. Well logs from six drill sites in Area C offshore India (Figure 1) were used to create a composite input data set serving as features in subsequent ML training. The selected features, extracted from well logs included bulk density (ρ), electrical resistivity (R_t), natural gamma radiation (GR), compressional wave velocity (V_p), and density-derived porosity (ϕ). The density-derived porosity was calculated using the standard density-porosity relationship (Eq. 1):

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_w} \quad (1)$$

where ρ_b is bulk density (g/cm³) measured by the density log, ρ_{ma} is the matrix density derived from the core analysis (Holland et al., 2019), and ρ_w is the formation water density with assumed value of 1.05 g/cm³.

Figures 2 and S2 provide example montages of well logs used as features in the ML training process together with RAB images from Site NGHP-02-07. In this study, S_{mh} and methane hydrate occurrence are the response variables of interest with the remaining features serving as input variables for trained models. Archie derived S_{mh} were used as the ground truth for assessing the accuracy of the ML approach. The formulas of the Archie electrical resistivity method to determine gas hydrate saturations in sediments are given in Supplementary Materials.

At each well log measured data point (acquired every 0.5 ft along the full length of the well), the selection of pore-filling or fracture-filling methane hydrate occurrence was assigned as the ground truth obtained through the interpretation of the RAB images (e.g. Collett et al., 2019). The determination of methane hydrate pore-filling or fracture-filling and non-hydrate-bearing were assigned based on the analysis of resistivity, RAB images, and V_p well logs (Collett et al., 2019).

Data preprocessing involved three steps starting with depth cutoffs, excluding depth intervals showing washouts, and then outlier removal. Depth cutoffs entails removing very deep and shallow data points that are outside of the gas hydrate stability zone (GHSZ). The shallow data points were removed due to the extreme unconsolidated nature of the sediments close to seafloor where the log readings were unreliable. That led to unphysically property values (see,

for example, NMR and density porosities ranging from 0.6 to near 1.0 within first 30 m below sea flow in Figure 2). Table 2 displays the original depth intervals and intervals after cutoff used for each well. Depth points which show missing log readings in the previously mentioned LWD data suite were also removed at this step. Washouts designate depth intervals where the hole size measured by the caliper log is larger than the drill bit (Ugborugbo and Rao, 2009). Washouts are enlarged portions of boreholes that are formed by removing the solid part of the borehole wall during drilling and/or circulation operations. Usually, washout sections of the borehole indicate the presence of soft or unconsolidated rocks and are inferred by having enlarged borehole caliper measurements relative to the size of the drill bit. To lessen the effect of unreliable measurements due to washouts, this study elected to remove well log measurements (i.e., depths) where the caliper sizes is in the upper five percentile (Figure S3) ranging from 9.13 to 9.90 inches depending on site. The last step involved scanning for outliers through the GLOSS algorithm (van Stein et al., 2016). Table 2 displays the final number of data points for each well after preprocessing. To illustrate the reduction in data before and after preprocessing the scatter figures showing S_{mh} vs depth and scatter cross-plots of density vs compressional velocity were collected in Figure S4. The data points removed after preprocessing were highlighted with different color.

Table 2

Relevant depth intervals and data points

Sites NGHP- 02-	Original depth interval (m)	Depth interval after cutoffs (m)	Number of depth points before preprocessing	Number of depth points after preprocessing
-05	2839 – 3407	2943 – 3353	3739	2350
-06	2665 – 3197	2773 – 3148	3493	2308
-07	1935 – 2300	1999 – 2173	2414	1063
-08	2190 – 2568	2270 – 2500	2495	1416

-09	2241 – 2513	2398 – 2513	1832	578
-10	1845 – 2210	1925 – 2070	2394	891

The TensorFlow ML library was used to construct models with Keras, which is a Python library for ANN models (Abadi et al., 2016; Géron, 2019). Keras enables a high-level approach to neural network construction and includes the backpropagation training algorithm. It accepts array-like input data and many hyperparameter options. The second step in this workflow was necessary to tune hyperparameters to ensure the reliability of trained models. Tuning is needed because hyperparameters are variables that remain static during the execution of the learning algorithm, and therefore, require prior optimization. There are several hyperparameters applicable to ANN models including the network topology, dropout rate, and learning rate. The general structure of ANN models is comprised of an input layer, two hidden layers with a dropout layer after each hidden layer, and finally an output layer. The tunable hyperparameter of the network topology is the number of nodes per layer. Dropout rates for each dropout layer were tuned independently. The choice of two hidden layers was based on previous work where ANN models were applied to predict methane hydrate saturations in permafrost-associated reservoirs (Chong et al., 2022). Following the results of that work (Chong et al., 2022), other hyperparameters assume an 80%/20% training/validation data split (other splitting schemes, such as 70/30, 60/40, 90/10, were studied and 80%/20% split was found to be optimal (Singh et al., 2020)), rectified linear unit (ReLU) activation functions was used for the hidden layers, batch size of 100, 500 epochs, and the Adam optimization algorithm. For the loss function and the mean squared error was used for saturation value prediction, while categorical cross entropy was used

for occurrence classification. The tuning process utilized the hyperband algorithm, an infinite “armed bandit” approach to select optimal hyperparameters (Li et al., 2016), and the Keras Tuner hyperparameter optimization framework (O’Malley et al., 2019). A grid search for hyperparameters is shown in Figure S5. Tuning was performed by using each log as a feature and then different WLC were applied with each feature scaled using minimum and maximum values. All five features (ρ , ϕ , R_t , GR , and V_p) were used for the tuning process. Table 3 lists optimized hypermeters, features, and well log information used in the study.

Table 3

Optimized hyperparameters, features, and wells utilized to predict occurrence and methane hydrate saturations in sediments. *ReLU* is rectified linear unit; *Softmax* is normalized exponential function; *Adam* is the stochastic gradient descent algorithm; *MSE* (mean square error) was used for the loss function; *GR*, *Rt*, ρ , f , V_p are gamma radiation, electrical resistivity, bulk density, and compressional wave velocity, respectively.

Parameters and features	Methane hydrate saturation prediction (Regression analysis)	Methane hydrate occurrence prediction (Classification analysis)
runs	100	100
training split	80%	80%
validation split	20%	20%
activation function	ReLU	ReLU
1 st hidden layer	150	50
2 nd hidden layer	75	100
final layer function	Linear	Softmax
dropout rate after 1 st layer	10%	10%
dropout rate after 2 nd layer	10%	10%
learning rate	0.0005	0.0025
batch size	100	100
epochs	500	350
optimizer	Adam	Adam
loss function	Mean squared error	Categorical cross entropy
scaling	min max	min max
well logs	GR , R_t , ρ , f , V_p	GR , R_t , ρ , f , V_p
NGHP-02 sites/holes	#5-#10	#5-#10

Although models can be effectively trained using all available features, WLC with fewer features can yield acceptable accurate models (> 0.70 is acceptable for generic ML models (Allwright, 2022)) as found in a previous study (Chong et al., 2022). The utilization of fewer

features in WLC expands potential applicability of a trained ML method to physical sites that do not have all five of the reference well logs or some depth portions of the logs are not available. A forward selection method was adopted in this study for determining optimal WLC. Starting with every single and pair WLC, the average coefficient of determination (R^2) was analyzed, and accuracy was balanced for one hundred model outcomes (realizations). R^2 was the accuracy metric for the gas hydrate saturation prediction as derived by Eq. 2,

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (2)$$

where y_i , $\hat{y}_i$, and $\bar{y}$ are the expected the ground truth S_{mh} , predicted S_{mh} , and averaged predicted S_{mh} .

Each realization was generated using a separate data split and training run meaning that within the 80%/20% split selection, data were changed every time a new model run started. That, combined with arbitrary initialized *weights* (contributions) for each layer in a ML model (Glorot and Statistics, n.d.) provided the prediction of S_{mh} values at a certain depth point differs across realizations. Thus, enabled accuracy distributions to be built for each WLC.

For classification models, confusion matrices were generated to analyze each model. Table 4 shows the setup of the confusion matrix for the “fracture-filling” class as “positive” using the multitype classification scheme. Similar matrices were used for the “pore-filling” and “no occurrence of methane hydrate” types.

Table 4
Confusion matrix in respect of the “fracture-filling” type as “positive”.*

		Predicted		
		none	fracture	pore
Truth	none	TN	FP	TN
	fracture	FN	TP	FN
	pore	TN	FP	TN

252 *True Positive (TP) defines how many positive class samples a ML model predicted correctly;
253 True Negative (TN) defines how many negative class samples a ML model predicted correctly;
254 False Positive (FP) defines how many positive class samples a ML model predicted incorrectly;
255 False Negative (FN) defines how many negative class samples a ML model predicted incorrectly.

256 The performance metrics utilized for classification analysis are as follows: *Accuracy*, *Precision*,
257 *Recall*, *F-score*. The definitions of those metrics are placed in Supplementary Materials.

258 In other words, *Precision* measures a portion of all positive occurrence type recognition
259 that was correct, while *Recall* means the fraction of a correct occurrence type that a ML model
260 identifies correctly. A specific issue of well log information included in the training process is that
261 most depth points correspond to sediments bearing no gas hydrate (86.16% out of the total
262 number of depth points). That means that most of the occurrence classification is expected as
263 the “none” class. Thus, the data should be considered imbalanced because one of the labels is
264 dominant misleading to attributing high overall accuracies. To correct this, a balanced accuracy
265 approach was used:

$$266 \text{ Balanced Accuracy} = \frac{1}{2} \left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP} \right) \quad (3)$$

267 Such solutions are routinely used in the classification analysis to assess performance of ML
268 algorithms (Tharwat, 2021).

269 **Methane Hydrate Occurrence Prediction**

270 To characterize marine gas hydrate systems the ML application workflow used in this study
271 consisted of two parts. The first part is a categorial analysis to predict the type of gas hydrate
272 occurrence (i.e., fracture-filling or pore-filling methane hydrate). To do so, ML models were
273 trained using WLC against RAB-interpreted assignments of pore-filling and fracture-filling
274 methane hydrate, considered as the ground truth (Collett et al., 2019). The purpose of the

categorical analysis was to identify a type of methane hydrate occurrence in accumulations and delineate methane hydrate-bearing intervals within a stratigraphic section of interest.

Table 5

Averaged balanced accuracy calculated from 100 realizations for all WLC of the features used to predict methane hydrate occurrence for all wells. Green color bar scales horizontally with averaged balanced accuracy of each WLC; standard deviation coloration scales from red (highest) to white (intermediate) to blue (lowest).

WLC					Averaged balanced accuracy	Standard deviation
GR	Rt	ρ	ϕ	Vp	0.8573	0.0283
GR	Rt	ρ		Vp	0.8550	0.0236
GR	Rt		ϕ	Vp	0.8547	0.0247
	Rt		ϕ	Vp	0.8431	0.0239
	Rt	ρ		Vp	0.8418	0.0278
	Rt	ρ	ϕ	Vp	0.8374	0.0241
GR	Rt			Vp	0.8173	0.0284
GR	Rt		ϕ		0.8038	0.0234
GR	Rt	ρ	ϕ		0.7990	0.0242
GR	Rt	ρ			0.7916	0.0237
	Rt			Vp	0.7908	0.0265
	Rt		ϕ		0.7715	0.0261
	Rt	ρ			0.7698	0.0238
	Rt	ρ	ϕ		0.7670	0.0320
GR	Rt				0.7309	0.0272
GR		ρ	ϕ	Vp	0.7241	0.0241
GR		ρ		Vp	0.7236	0.0247
GR			ϕ	Vp	0.7170	0.0285
	Rt				0.6785	0.0323
		ρ	ϕ	Vp	0.6613	0.0252
				Vp	0.6602	0.0312
	Rt			Vp	0.6548	0.0300
GR		ρ		Vp	0.6432	0.0304
GR			ϕ		0.5511	0.0192
GR		ρ			0.5503	0.0219
GR		ρ	ϕ		0.5471	0.0192
				Vp	0.4792	0.0190
		ρ			0.3334	0.0012
GR					0.3333	0.0000
			ϕ		0.3333	0.0001
		ρ	ϕ		0.3333	0.0001

Table 5 lists the ranking of all WLC in descending order of the average balanced accuracy for 100 runs of the model. The highest accuracy corresponds to the WLC including all features. As the number of features in a WLC decreases there is a general trend of decreased accuracy. However, the first ten WLCs listed in Table 5 with a reduction of the number of features to four and three still show accuracy > 0.79 , and the top seven WLC results are within a single standard deviation. These results are encouraging because the observed relatively higher accuracies with fewer well logs will allow methane hydrate occurrence prediction at more sites including those with few well logs available. Moreover, all seven of the top WLC include R_t and V_p . Thus, it is highlighting the importance of R_t and V_p as accurate predictors of methane hydrate occurrence, which is consistent with the more traditional physics-driven methods for estimating gas hydrate saturations in sediments (Collett and Lee, 2012; Lee et al., 1996).

Figure 3 displays results for the best performing WLC in terms of balanced accuracy, which is a metric well-suited for these data since most depth points return a none (no gas hydrate) type as true. The figure also displays the categorical accuracy reflecting the fraction of positively identified pore- and fracture-filling among true data. Figure 3 also provides an assessment of the model performance showing the balanced accuracy (0.8573 ± 0.0283) decreased compared to the categorical accuracy (0.9609 ± 0.0053). The broad distribution of balanced accuracy is related to sensitivity of that accuracy to proportions of pore- and fracture-filling types within the 80/20 split for every realization. For categorical accuracy the unbalanced data are dominated with the none type that ML models tend to predict more accurately leading to a narrow distribution for

all realizations (Figure 3). Other performance metrics like precision, recall, and F-score were also calculated (Figure 4).

Analysis of the performance metric distributions shows that there is high fidelity in the prediction of the none classification types that dominate the dataset, while the remaining two types of methane hydrate occurrence reveal broad ranges of metrics. For the precision metric, the range is within 0.750 - 0.990 for both fracture- and pore-filling methane hydrate types with averaged values equal to 0.9017 ± 0.0485 and 0.8412 ± 0.0417 , respectively. Therefore, better precision is achieved for the fracture-filling compared to pore-filling methane hydrate occurrence. The same trend is preserved for the recall metric with more pronounced difference in averaged numbers, 0.8511 ± 0.0568 and 0.7036 ± 0.0553 , for fracture- and pore-filling, respectively. In other words, precision and recall provide good predictions for the fracture-filling type, while pore-filling type predictability is worse, especially for the recall metric. The better performance in prediction of the fracture-filling type is further manifested in the F-score metric that reports a harmonic mean of precision and recall (Figure 4) with averaged values calculated to be 0.8907 ± 0.0250 and 0.7644 ± 0.0342 , for fracture-filling and pore-filling, respectively.

Tables S1-S6 includes the performance metrics determined for each well examined in this study to differentiate a relative contribution of the model predictions. For the pore-filling type, the precision numbers are noticeably and consistently better than recall and vary from 0.807 to 0.999. The recall numbers range from 0.586 to 0.871. Fracture-filling type shows improved quality from 0.920 to 0.957 and from 0.888 to 0.902 for precision and recall, respectively. In other words, based on the definitions of precision and recall, a ML model performs better in correctly

predicting the type of occurrence at every depth points compared to model performance in correctly predicting the actual occurrence type within corresponding depth intervals (a proportion of actual positives identified correctly). Precision and recall seem to be equally important in the occurrence prediction, so F-score incorporating both terms should be considered in the evaluation of the performance of the ML models.

The importance of identifying occurrence type is related to a fact that pore-filling gas hydrate reservoirs that typically reside in sandy-rich reservoirs with elevated porosity and intrinsic permeabilities (Wang et al., 2016) are potentially suitable for future production testing. From another perspective, methane hydrate found in fractures and veins often exhibits low saturation values in mud- and clay-rich sediments with low porosities and low intrinsic permeabilities (Zhao et al., 2022). Accumulations in mud and clay-rich sediments rarely present attractive resource compared to deposits found in sand-rich reservoir systems (Fujii et al., 2015; Max and Johnson, 2019; Zhao et al., 2022). Thus, by utilizing ML models trained with WLC, it is possible to identify gas hydrate reservoirs within continuous geological sedimentary intervals (typically ranging from about 5 to 20 m thick), which can also be used to identify specific types of gas hydrate occurrences. It should be noted that gas hydrate accumulations, especially in marine environment often exhibit finely interbedded depositional properties where sandy layers with high gas hydrate saturations alternate with muddy layers with little to no methane hydrate (Fujii et al., 2015). These interbedded methane hydrate occurrences pose a challenge when it comes to acquiring accurate log-derived petrophysical measurements when the thicknesses of mud-

sand alternating layers are below the logging tools capability to resolve each of the relatively thin interbeds (Boswell et al., 2019).

Methane Hydrate Saturation Prediction

After delineating gas hydrate reservoir boundaries and identifying dominant type of methane hydrate occurrence, the next part of the workflow involved a regression analysis estimation of S_{mh} . Table 6 includes the ranking of all WLC relative to their ability to predict S_{mh} values in descending order of their averaged R^2 values. The top ten places in Table 6 correspond to the WLC including 3–5 features that yielded accuracies within the range of 0.720–0.760, with a general trend to slightly decreasing R^2 values with the reduction in the number of features in WLC. For the first ten WLC, the referenced features are similar to those that yield the best performers for the analysis of the types of the methane hydrate occurrence.

Using R_t (i.e., resistivity logs) alone yields poor performance ($R^2 = 0.619$), even though it is the primary input into the reference Archie relationship with empirical parameters. It should be noted the hyperparameters of ANN include two hidden layers (Table 3) that provide non-linear dependency between a feature (R_t) and a target saturation value (S_{mh}) in this data-driven approach, which is consistent with non-linearity of the Archie model. The ML model requires R_t to be augmented with the other features like V_p and GR to improve the methane hydrate saturation predictions. As R_t and V_p are capable of recording evidence for the occurrence of gas hydrate, it is expected that a combination of such features in an ML model would perform better when compared to a standalone R_t or V_p derived values.

Tables 5 and 6 demonstrate that WLC assessment can be made to achieve high overall accuracy for the determination of methane hydrate occurrence and saturation predictions. The top three, as posted in Tables 5 and 6, for both rankings share GR, Rt, and Vp as common WLC with a fourth well log being either density or porosity that slightly improves R^2 or balanced accuracy. For the models with only three WLC, the highest performing WLC is Rt, porosity, and Vp. Thus, for future reference in developing gas hydrate-related models, using well logs that contain a combination of at least Rt and Vp would be desirable features in addition to GR or porosity.

The NGHP-02 wells studied in this work penetrated sections with no methane hydrate or either pore-filling methane hydrate or fracture-filling depositions (Table 1). The spacing between the wells used in the study are highly variable (Figure 1), with Sites NGHP-02-05 and -6 located in deep water and showing geological architecture distinct from the remaining sites (Collett et al., 2019). In this regard, these sites are analyzed separately from the other. Figure 5 depicts R^2 distributions collected for each site and overall distributions for all the wells. The R^2 values range from 0.680 to 0.920 with a tendency in the distributions to become narrower for wells with better predictions and more consistency due to fewer data points (like Site NGHP-02-09 and -10). This observation is consistent with early studies for the prediction of S_{mh} for methane hydrate accumulations on the Alaska North Slope and in northern Canada, where R^2 distributions with better accuracies also demonstrates better precision (Chong et al., 2022). Figure 5 also illustrates the decreasing accuracies for the predictions for Sites NGHP-02-05, -06, and -08 compared to Sites NGHP-02-09 and -10.

Table 6

Averaged coefficient of determination (R^2) values calculated from 100 realizations for all WLC of the features used to predict gas hydrate saturations for all wells. Green color bar scales horizontally with averaged balanced accuracy of each WLC; standard deviation coloration scales from red (highest) to white (intermediate) to blue (lowest).

WLC				R ² avg	R ² stdev
GR	Rt	ρ	Vp	0.7603	0.0272
GR	Rt	ρ	ϕ Vp	0.7602	0.0285
GR	Rt	ϕ	Vp	0.7579	0.0275
	Rt	ρ	ϕ Vp	0.7529	0.0336
	Rt	ϕ	Vp	0.7497	0.0286
	Rt	ρ	Vp	0.7441	0.0290
GR	Rt	ρ	ϕ	0.7255	0.0289
GR	Rt	ϕ		0.7242	0.0286
GR	Rt		Vp	0.7220	0.0312
GR	Rt	ρ		0.7198	0.0319
	Rt	ϕ		0.7086	0.0289
	Rt	ρ		0.7081	0.0326
	Rt	ρ	ϕ	0.7003	0.0323
	Rt		Vp	0.6990	0.0333
GR	Rt			0.6880	0.0294
	Rt			0.6193	0.0342
GR		ρ	Vp	0.5266	0.0382
GR		ϕ	Vp	0.5258	0.0346
GR		ρ	ϕ Vp	0.5225	0.0344
		ρ	ϕ Vp	0.4456	0.0332
		ϕ	Vp	0.4357	0.0385
		ρ	Vp	0.4317	0.0346
GR			Vp	0.4218	0.0386
			Vp	0.3031	0.0337
GR		ϕ		0.2617	0.0346
GR		ρ	ϕ	0.2595	0.0295
GR		ρ		0.2557	0.0314
GR				0.1265	0.0156
		ρ	ϕ	0.0550	0.0161
		ϕ		0.0543	0.0158
		ρ		0.0515	0.0150

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Figure 6 plots predicted and expected S_{mh} numbers for two sites. The S_{mh} prediction for Site NGHP-02-05 shows multiple small peaks, with limited thickness, that are not confirmed by the ground truth data and that contribute to decreased performance. A similar trend is found for Site NGHP-02-06, which demonstrates the lowest R^2 averaged value (Figure 5). Sites NGHP-02-05 and -06 sit in the deepwater of Area C in the Krishna-Godavari Basin (Figure 1) which is interpreted to represent a distal stacked fan system in GHSZ (Collett et al., 2019). These sites are characterized by prominent interbedded sedimentary sections and are interpreted to occur within a “gas-poor” system with only limited gas charge and partially gas hydrate filled reservoir sections (Collett et al., 2019). In other words, the sediment properties allow for higher S_{mh} than actually occurs in the pore space. This can contribute to decreased ML performance for those wells as the learning algorithm based on porosity and natural gamma ray logs in WLC expects higher saturation values similar to the other wells.

In previous work involving ML prediction of S_{mh} in methane hydrate accumulations on the Alaska North Slope and in northern Canada, accuracies varied between 80–90% ($R^2 = 0.80$ – 0.90) estimated against a ground truth derived from NMR-based techniques (Chong et al., 2022). These numbers demonstrate a higher degree of accuracy for the terrestrial Arctic deposits than for the most part represent continuous and thick reservoir sections (Chong et al., 2022). On the other hand, the offshore India sites considered in this study are characterized by a thinly interbedded architecture (typical for marine systems) where sandy-rich gas hydrate-bearing units are interbedded with water-saturated (i.e., no gas hydrate) clayey shale layers (Figure S6 for Site NGHP-02-09 (Moridis et al., 2019)). The thicknesses of units in these thinly-interbedded

sequences approach the vertical resolution of LWD tools used to acquire well logs (Table 2 (Collett et al., 2019)). As a result, increased uncertainty in well log measurements exists where bed thickness falls below the vertical resolution of LWD tools and wireline devices used to acquire the downhole log data (Figure 2 in (Boswell et al., 2017)). There is an apparent difference between the terrestrial Arctic gas hydrate accumulations that typically occurs within a thick, continuous reservoir sections and marine deposits where thin hydrate-bearing layers often alternate with thin mud sub-units. This difference most likely contributes to the better ML performance for the former accumulations.

Predicted *versus* expected (the ground truth) saturation values reveal a broad distribution in values ranging from 0% to 92%, with a scatter of data points that to the right and below the 1:1 line as projected in Figs. 7 and 8. The data in Figure 7 suggest that a portion of the predicted values systematically underestimate the expected ground truth data, especially for Sites NGHP-02-06 and -08. The poor performance of the ML models for Site NGHP-02-06 is expected given that Sites NGHP-02-05 and -06 are found to be in gas deficient areas (Collett et al., 2019) leading to lower than potentially possible gas hydrate saturations, and the wells sit in the deepwater of the basin with distinctly different stratigraphic relationships compared to the other sites in this study. What caused the poor performance of the model predictions for Site NGHP-02-08 remains unclear. It is possible that the tool resolution issue related to the thinly-interbedded nature of gas hydrate accumulations and the fidelity of well logs may have contributed to the apparent less accurate predicted versus expected performance for Site NGHP-02-08. Although, based on the nature of the physical occurrence of methane hydrate at Site NGHP-02-08 characterized by more

massively bedded pore-filling and fracture sections dominated methane hydrate-bearing sections (Collett et al., 2019), it is likely that the presence of the fracture-dominated methane hydrate is the reason for the poor model performance. The Archie resistivity log predicted S_{mh} values for fracture-filling reservoirs have been shown to systematically underestimate the actual S_{mh} values (Figure 8). This requires additional features to be used with R_t to achieve a higher R^2 in the top ranking WLCs in fracture-dominated gas hydrate systems. The ML values that predicted lower S_{mh} values than the ground truth values for fracture-filling gas hydrate systems is considered a positive result since the ground truth tends to overestimate the actual values (Archie, 1942).

The uniform shift of the data points as plotted in both Figs. 7 and 8 contrasts with previous work predicting gas hydrate saturation in terrestrial Arctic gas hydrate deposits (Chong et al., 2022), where predicted and expected values were similar for gas hydrate saturations above 50% and saturations were underestimated when ground truth values were below 50%. In that work, it has been suggested that the systematical underestimation of S_{mh} values as predicted by ML for of methane hydrate intervals characterized with S_{mh} below 50% could be a shortcoming of the NMR-density porosity method used to estimate S_{mh} with the NMR-density method being sensitive to the presence of shale (particularly clay content) (Chong et al., 2022). In this study, the ground truth was estimated from the resistivity logs indicating the presence of methane hydrate in deposits that acts as an electrical insulator, and the recorded signal was less dependent on the compositional makeup of the host matrix (although still dependent on the host sediment petrophysical properties such as porosity and fluid content). Consequently, in this study

there is little-to-no sensitivity of the ML predicted values with respect to the relative range of expected saturation values.

Another important observation is associated with R_t logs that may not provide representative resistivity log values in sedimentary sections with anisotropic properties, such as gas hydrate filled fractures (Archie, 1942). These sections are usually enriched with clay, characterized with low porosity and low permeability values, and low S_{mh} compared to sandy-rich sediments with pore-filling gas hydrate (Collett et al., 2019). Such deposits rarely represent attractive targets for exploration (Boswell et al., 2014; Collett et al., 2019, 2011). Since in this study, we rely on the Archie relationship utilizing R_t log values measured in fracture-filling gas hydrate sections for deriving the ground truth, potentially poor measurements of anisotropic electrical properties are included into the Archie equation. That could lead to overestimation of S_{mh} for fracture-filling gas hydrate by up 50% compared to S_{mh} evaluated from core samples (Lee and Collett, 2009). Consequently, resistivity log-derived S_{mh} values for fracture-filling methane hydrate systems may overestimate the actual S_{mh} values.

The ML models trained and tested here provide the assessment of gas hydrate occurrences as well as quantitative assessment of gas hydrate saturation. To further improve the trained ML models created in this study, log data from the other sites in the offshore India, or another basin like the eastern Nankai Trough offshore Japan (Yamamoto et al., 2014) could be used to further train the developed ML models. Information from additional sources would also represent an excellent opportunity to expand the training datasets and validate the ML models using log data from sites not included in the training workflow.

Conclusion

In this study, ML models were developed using the well log data collected at six sites (from Sites NGHP-02-05, -06, -07, -08, -09, and -10) located in the Krishna-Godavari Basin in the offshore India (Area C) to predict the physical nature of methane hydrate occurrence and sediment void-space methane hydrate saturations. The categorical analysis using ML-trained models based on combinations of three to five well logs provided good predictions of the physical presence of methane hydrate relative to the ground truth determinations, yielding balanced accuracies from 0.79 to 0.86. The precision of the predictions was reported ranging from 0.74 to 0.99 and averaging 0.84 and 0.90 for the fracture-filling and the pore-filling occurrences, respectively. The recall was reported ranging from 0.59 to 0.99 and averaging 0.85 and 0.70 for the fracture-filling and the pore-filling occurrences, respectively. For the regression ML model predicted methane hydrate saturations at the six sites, the R^2 derived accuracy distributions ranged from 0.68 to 0.92.

Overall, the application of ML to predict gas hydrate occurrences and saturations revealed good performance against the ground truth measurements. Considering previous success in predicting methane hydrate saturations on Alaska North Slope and in northern Canada (Chong et al., 2022), results from this work indicate the viability of data-driven methods as alternative (or supplementary) to conventional physics-driven methods. Reliable and reproducible predictions of reservoir characteristics are critical not only for reservoir production performance studies, but also for gas hydrate reservoir genesis and system evolution from a scientific and exploration perspective. Often specialized log data (like NMR-logs) along with general well log

data (density and natural gamma-ray) are collected to target specific values for estimations of key reservoir parameters. On the other hand, more commonly acquired well logs can be utilized by ML to extract valuable information, possibly without running more specialized logs. Another promising application of ML is to reuse legacy data to assess emerging demands. Trained ML models may be able to deliver information using a relatively limited collection of well logs obtained in a legacy well. While commonly accepted in oil and gas industry, the ML models, such as the one presented in this study, have not widely used in the gas hydrate research community. This study shows the application of ML approach to yield meaningful information on the nature of gas hydrate reservoirs. The application of ML models can be further expanded, for instance, for studies on global scale environmental impacts of gas hydrate when combined with basin scale modeling.

Data availability

Data associated with this research are available and can be obtained by contacting the corresponding author.

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The list of figure captions:

Figure 1. A) National Gas Hydrate Expedition 01 (NGHP-01) and NGHP-02 site maps showing the site locations of wells drilled during the expeditions in the Krishna-Godavari (Area B, Area C, and Area E) and Mahanadi (Area A) offshore areas of India, **B)** Expanded view of Area C and Area E showing names of individual sites (adopted from Collett et al., 2019).

Figure 2. Examples of fracture-filling (75-103 mbsf) and pore-filling (246-271 mbsf) gas hydrate occurrences as interpreted from downhole acquired resistivity logs (P40H measured in ohm-m) and resistivity at bit image log (RAB; Deep resistivity static image) from Hole NGHP-02-08A (modified from Collett et al., 2019).

Figure 3. Comparison of the balanced and categorical accuracy distributions of methane hydrate occurrence assignment for the Sites NGHP-02-05, -06, -07, -08, -09 and -10 using the best performing WLC.

Figure 4. Precision, recall, and F-score value distributions for the methane hydrate occurrence assignment for all the NGHP-02 sites.

Figure 5. Average coefficient of determination (R^2) distributions generated using the ML models with WLC including GR, Rt, density, and Vp features (see ranking listed in Table 6).

Figure 6. Methane hydrate saturation predictions for Site NGHP-02-05 (top) and Site NGHP-02-09 (bottom) with depth using the ML models with WLC including GR, Rt, density, and Vp features (see ranking listed in Table 6).

Figure 7. Prediction versus expectation of S_{mh} values depicted as a scatter plot created for pore-filling reservoir sections only using the ML models with GR, Rt, density, and Vp features (see ranking listed in Table 6). Straight line indicates a 1:1 match for pore-filling reservoir sections.

Figure 8. Prediction versus expectation of S_{mh} values depicted as a scatter plot created for fracture-filling reservoir sections only using the ML models with GR, Rt, density, and Vp features (see ranking listed in Table 6). Straight line indicates a 1:1 match for fracture-filling reservoir sections.

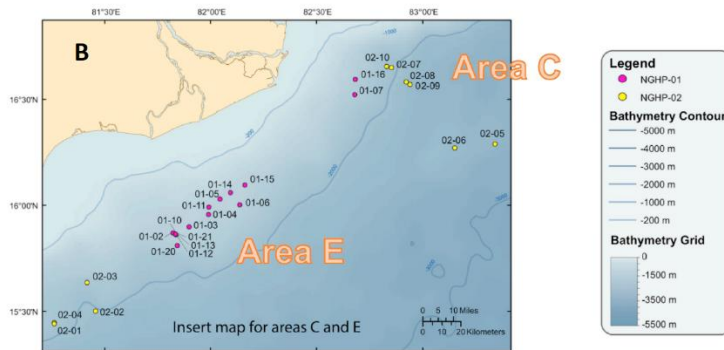
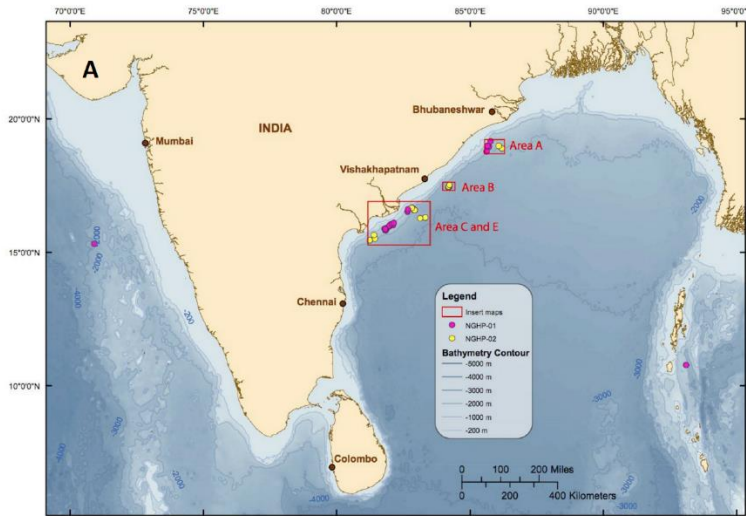
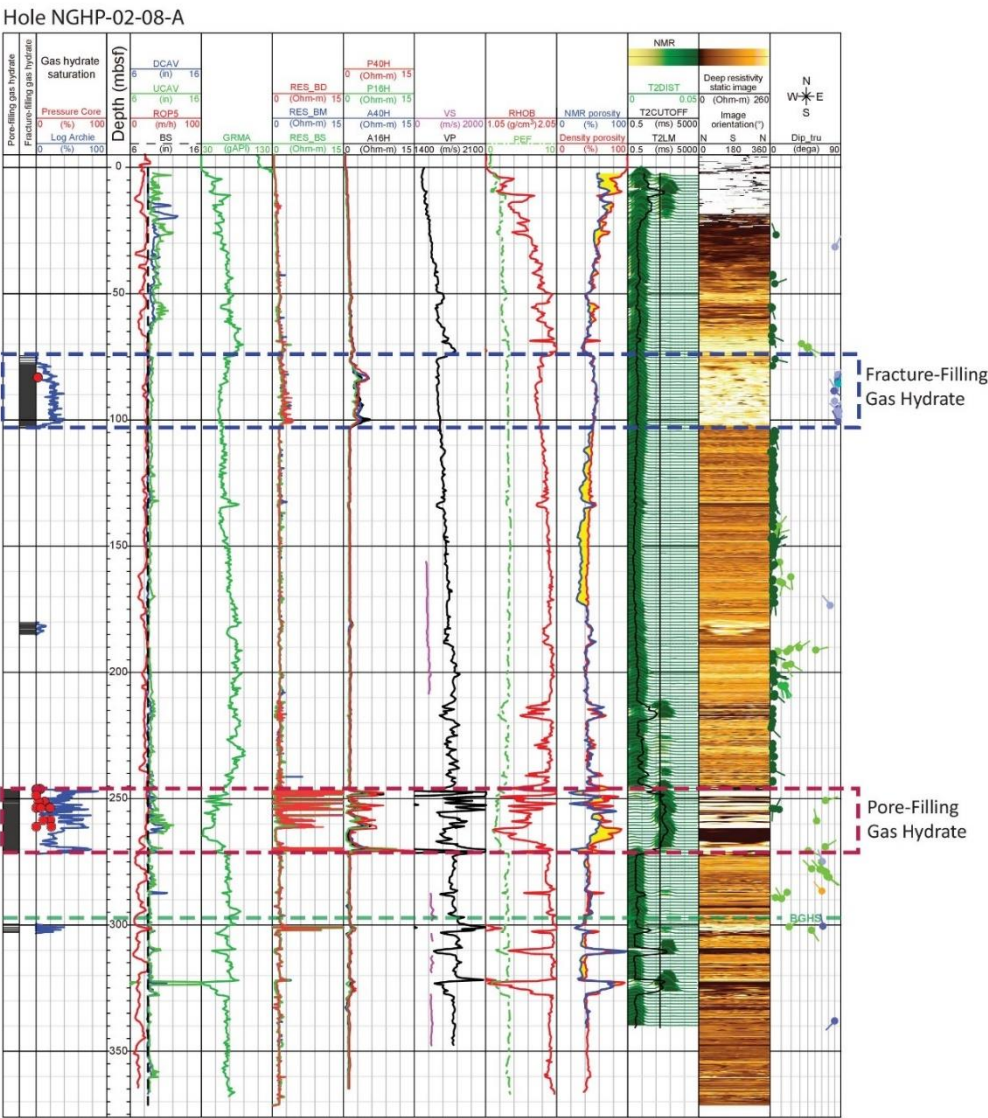
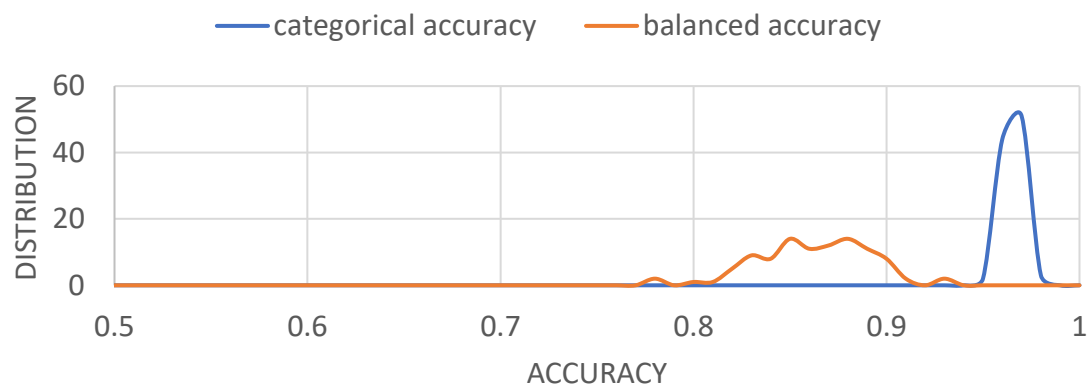


Figure 1. A) National Gas Hydrate Expedition 01 (NGHP-01) and NGHP-02 site maps showing the site locations of wells drilled during the expeditions in the Krishna-Godavari (Area B, Area C, and Area E) and Mahanadi (Area A) offshore areas of India, **B)** Expanded view of Area C and Area E showing names of individual sites (adopted from Collett et al., 2019).



715 **Figure 2.** Examples of fracture-filling (75-103 mbsf) and pore-filling (246-271 mbsf) gas hydrate occurrences as
716 interpreted from downhole acquired resistivity logs (P40H measured in ohm-m) and resistivity at bit image log (RAB;
717 Deep resistivity static image) from Hole NGHP-02-08A (modified from Collett et al., 2019).
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Figure 3. Comparison of the balanced and categorical accuracy distributions of methane hydrate occurrence assignment for the Sites NGHP-02-05, -06, -07, -08, -09 and -10 using the best performing WLC.

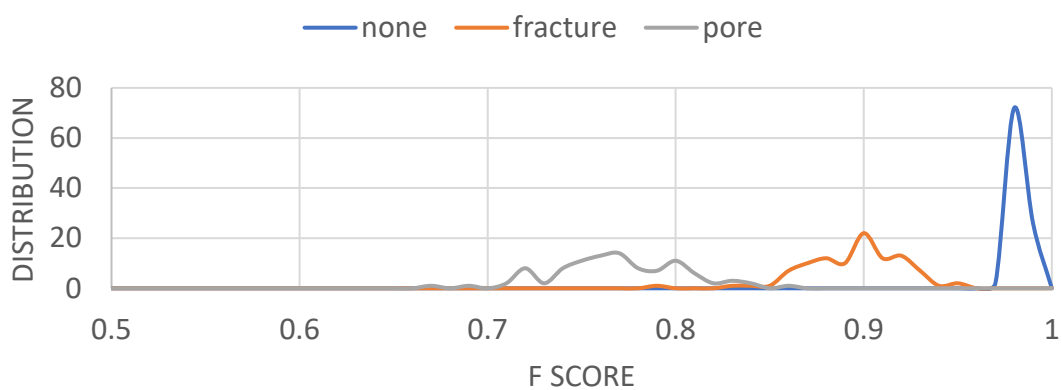
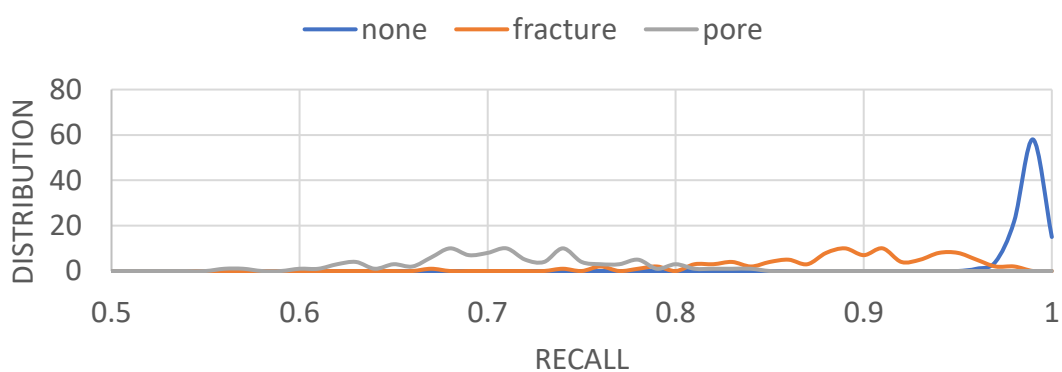
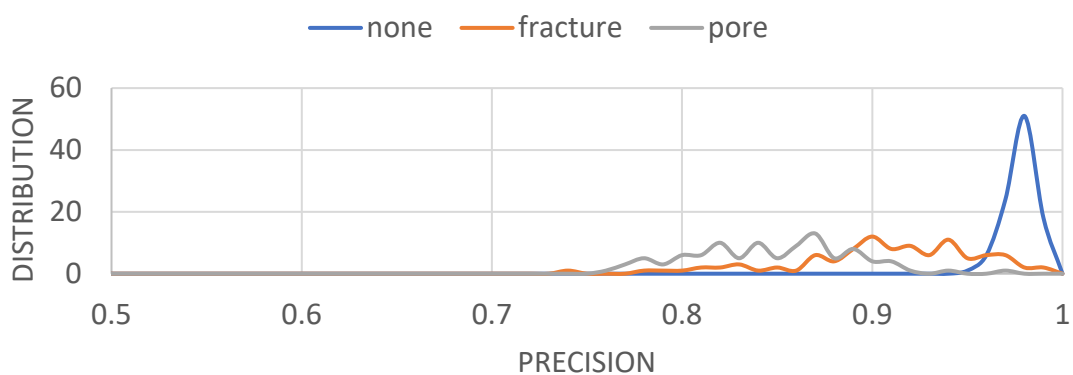
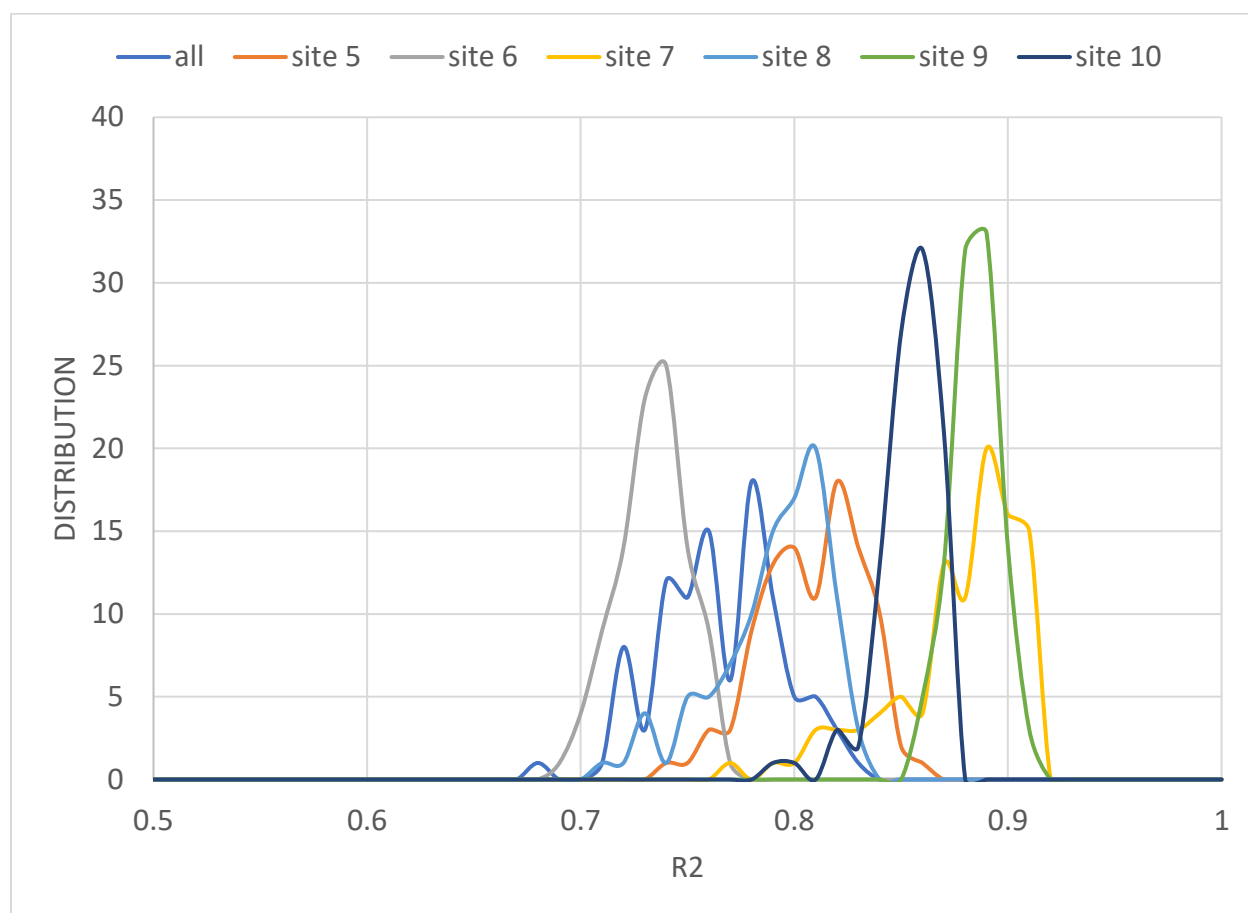


Figure 4. Precision, recall, and F-score value distributions for the methane hydrate occurrence assignment for all the NGHP-02 sites.

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733 **Figure 5.** Average coefficient of determination (R^2) distributions generated using the ML models with WLC including
 734 GR, Rt ,density, and Vp features (see ranking listed in Table 6).
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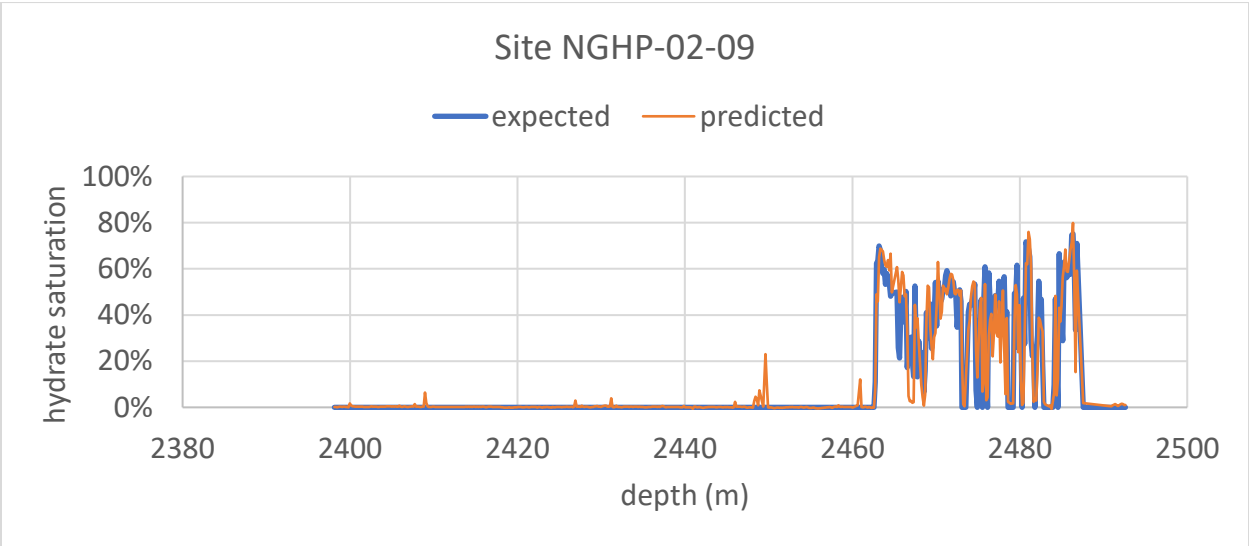
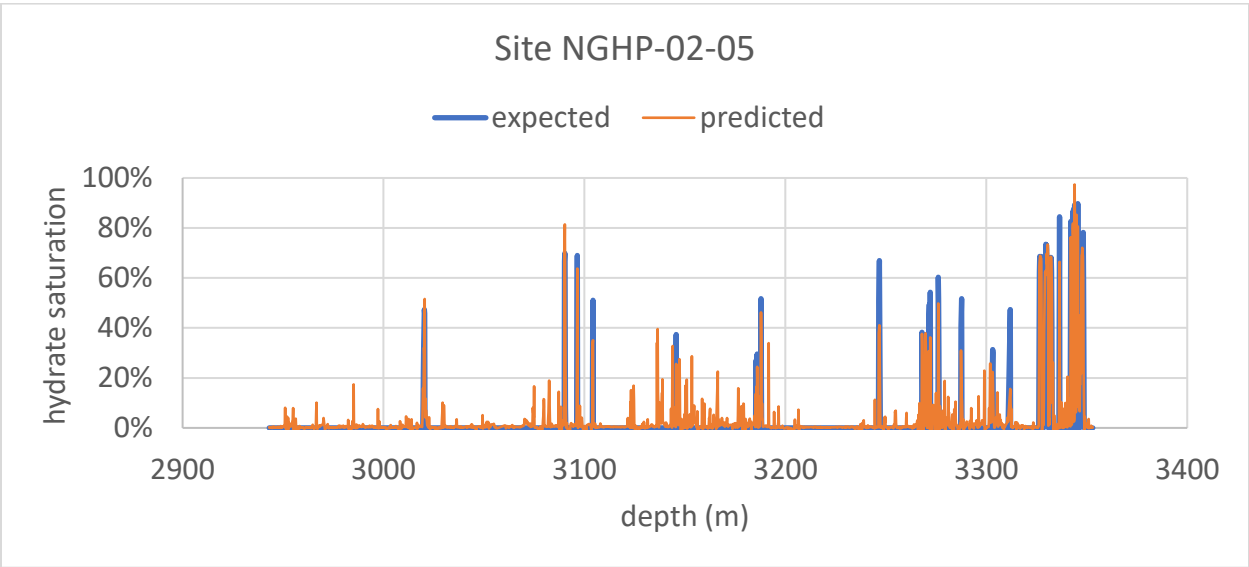


Figure 6. Methane hydrate saturation predictions for Site NGHP-02-05 (top) and Site NGHP-02-09 (bottom) with depth using the ML models with WLC including GR, Rt, density, and Vp features (see ranking listed in Table 6).

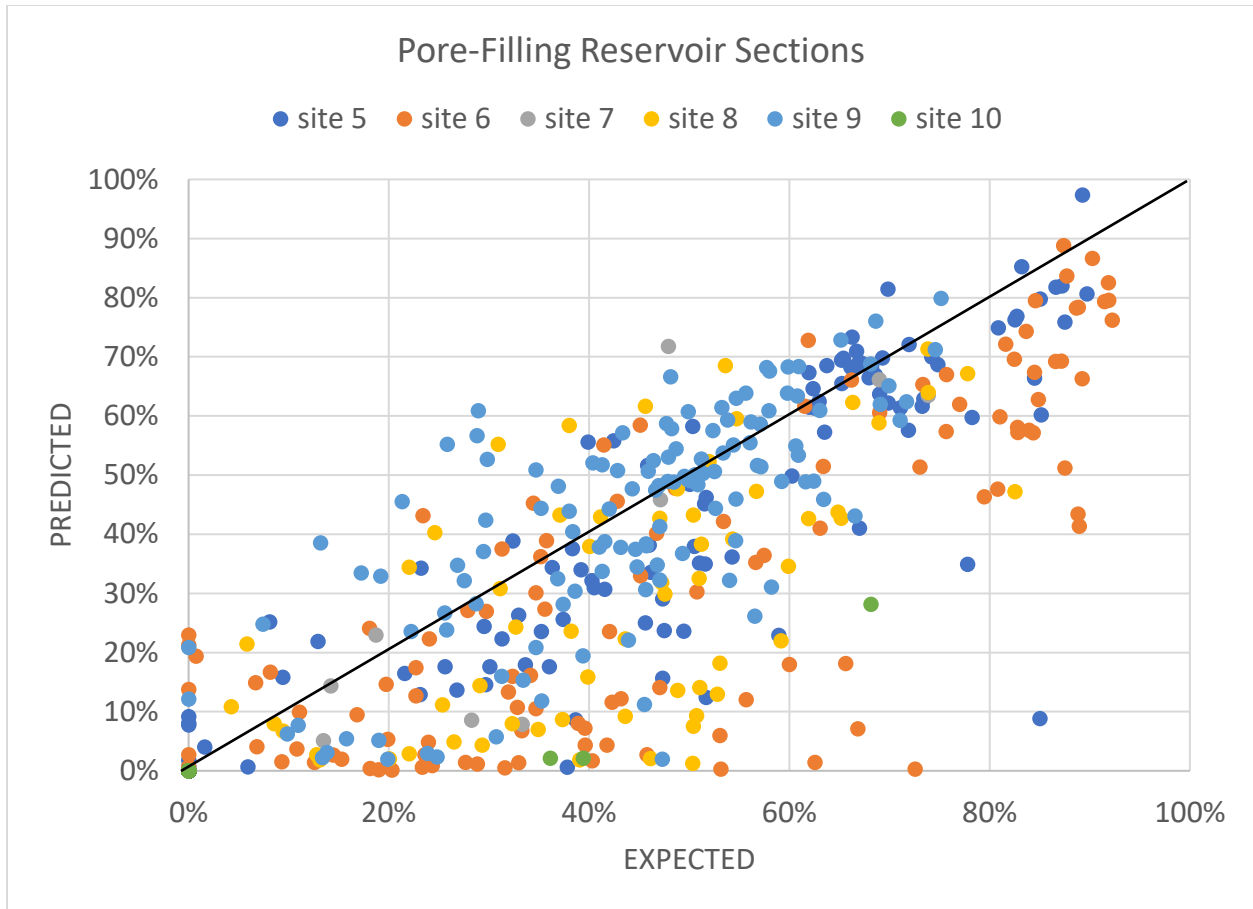


Figure 7. Prediction versus expectation of S_{mh} values depicted as a scatter plot created for pore-filling reservoir sections only using the ML models with GR, Rt, density, and Vp features (see ranking listed in Table 6). Straight line indicates a 1:1 match for pore-filling reservoir sections.

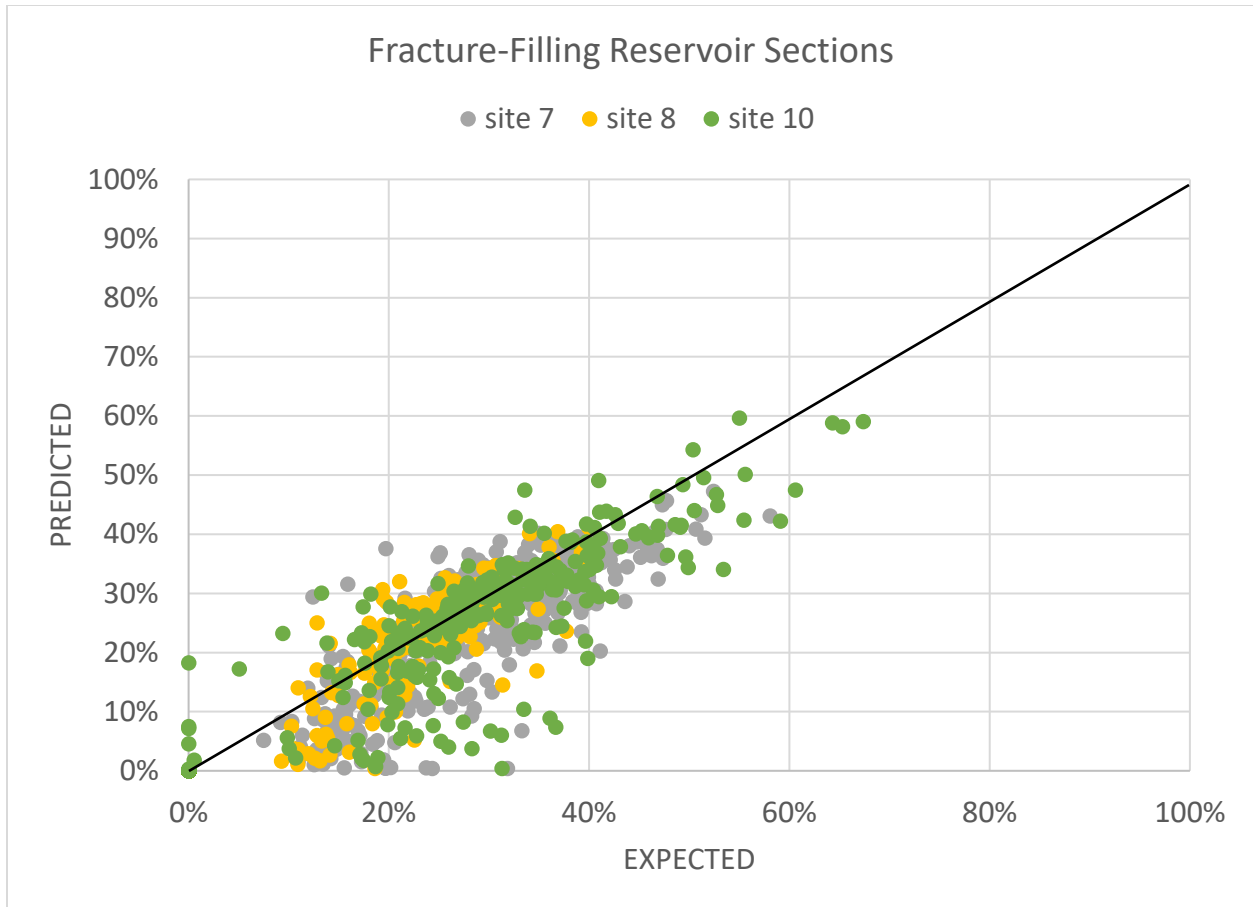


Figure 8. Prediction versus expectation of S_{mh} values depicted as a scatter plot created for fracture-filling reservoir sections only using the ML models with GR, Rt, density, and Vp features (see ranking listed in Table 6). Straight line indicates a 1:1 match for fracture-filling reservoir sections.